

RECAP LINEAR ALGEBRA

- m EQUALITIES in m VARIABLES (SYSTEM)

$$\Rightarrow Cx = d \quad \begin{array}{l} C \text{ } m \times m \text{ matrix} \\ x \text{ } m \text{ column vector} \\ d \text{ } m \text{ column vector} \end{array}$$

- NECESSARY and SUFFICIENT CONDITIONS to have a UNIQUE SOL

$$\text{row} \cdot \text{rank}(C) = \text{column} \cdot \text{rank}(C) = m$$

All the rows of C are LINEARLY INDEPENDENT

All the columns of C are LINEARLY INDEPENDENT

C is INVERTIBLE, C^{-1} EXISTS ($\det C \neq 0$)

All these conditions are equivalent (if one holds also the others do).

- If these conditions do NOT hold, the system is either IMPOSSIBLE or MANY SOLUTIONS exist

EXAMPLE:

$$\begin{array}{ll} \max & Z = 30x_1 + 50x_2 \\ \text{st} & 4x_1 + 11x_2 \leq 880 \\ & x_1 + x_2 \leq 150 \\ & x_2 \leq 60 \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

How to go
from EQ to
MATRICES

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad d = \begin{bmatrix} 880 \\ 150 \\ 60 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 11 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{array}{ll} \max & Z = 30x_1 + 50x_2 \\ \text{st} & Cx \leq d \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

(note that in this case we have inequalities and we can have different dimensions - C is 3×2)

LINEAR ALGEBRA DEFINITIONS

LINEAR COMBINATION: given $v_1, \dots, v_n \in V$, a vector $v \in V$ is a linear combination of $v_1, \dots, v_n$ if there exist $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ such that

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n$$

LINEAR DEPENDENT: $v_1, \dots, v_n \in V$ are linearly dependent if there exist $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ such that at least one $\alpha_i \neq 0$ and

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0$$

LINEAR INDEPENDENT: if they are not dependent, i.e., if

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0$$

have $(\alpha_1, \dots, \alpha_n) = 0$ as the only solution

BASIS: a set of vectors $v_1, \dots, v_n \in V$ is called a basis if the linear space V is generated by $v_1, \dots, v_n$ (i.e., every vector in V is a linear combination of $v_1, \dots, v_n$) and $v_1, \dots, v_n$ are linearly independent

RANK: number of linearly independent rows
number of linearly independent columns